

EDU OS Scheduler System

Project Overview

EDU OS Scheduler System is a hybrid Operating System simulation project developed using C and Python.

The project demonstrates:

- CPU scheduling algorithms
- Process management
- Thread synchronization
- IPC concepts
- Deadlock simulation
- Shared memory concepts
- Scheduling visualization
- Metrics analysis

The system combines:

- Low-level OS concepts in C
 - Visualization and analytics in Python
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Features

Python Scheduler System

Implemented algorithms:

- FCFS Scheduling
- SJF Scheduling
- Priority Scheduling
- Round Robin Scheduling

Additional features:

- Gantt Chart Visualization
 - Scheduling Metrics
 - Comparison Dashboard
 - Process Loader
 - Interactive Menu System
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C Scheduler System

Implemented modules:

- FCFS Scheduler
 - SJF Scheduler
 - Round Robin Scheduler
 - Priority Scheduler
 - Thread Demonstration
 - Producer Consumer IPC
 - Deadlock Detection
 - Shared Memory Demo
 - Mutex Synchronization Demo
 - PCB Serialization
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Technologies Used

Programming Languages

- C
- Python

Python Libraries

- matplotlib
- json

Tools

- GCC Compiler
 - Git
 - GitHub
 - Visual Studio Code
-

Project Structure

```
Edu_OS-23511351006/  
|  
├─ c_core/  
|   └─ eduos.exe  
|   └─ fcfs.c
```

```
|   |   | sjf.c
|   |   | rr.c
|   |   | main_sim.c
|   |   | scheduler.h
|   |   | └─ pcb_serializer.c
|
|   └─ python_scheduler/
|       |   | scheduler_sim.py
|       |   | gantt.py
|       |   | metrics.py
|       |   | dashboard.py
|       |   | loader.py
|       |   | └─ algorithms/
|       |       |   | fcfs.py
|       |       |   | sjf.py
|       |       |   | priority.py
|       |       |   | └─ round_robin.py
|
|   └─ controller/
|       |   | gantt_chart.py
|       |   | └─ main_controller.py
|
|   └─ docs/
|   └─ logs/
|   └─ tests/
|   └─ run_all.bat
|   └─ README.md
|   └─ pcb_snapshot.json
|   └─ simulation_report.json
```

Installation

Install Python Dependencies

Run:

```
pip install matplotlib
```

How To Run The Project

Run Python Scheduler

```
cd python_scheduler  
python scheduler_sim.py
```

Run C Scheduler

```
cd c_core  
.\veduos.exe
```

Run Entire System

Use:

```
.\run_all.bat
```

Scheduling Algorithms

FCFS

First Come First Serve executes processes in arrival order.

Characteristics:

- Simple
 - Non-preemptive
 - High waiting time possible
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SJF

Shortest Job First executes smallest burst time first.

Characteristics:

- Efficient average waiting time
 - Non-preemptive
 - Starvation possible
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Priority Scheduling

Processes execute according to priority value.

Characteristics:

- Important tasks execute first
 - Starvation possible
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Round Robin

Processes execute using a fixed time quantum.

Characteristics:

- Fair CPU allocation
 - Preemptive scheduling
 - Good for multitasking systems
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Metrics Calculated

The project calculates:

- Average Waiting Time
 - Average Turnaround Time
 - Throughput
 - CPU Utilization
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Visualization

The project supports:

- Console Gantt Charts

- Graphical Gantt Charts
 - Dashboard Comparisons
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Example Output

FCFS SCHEDULING

P1		WT=0		TAT=6
P2		WT=6		TAT=9
P3		WT=9		TAT=17

Future Improvements

Possible upgrades:

- Arrival time support
 - Preemptive SJF
 - Real-time simulation
 - GUI interface
 - Advanced deadlock handling
 - JSON process import/export
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Learning Outcomes

This project demonstrates understanding of:

- CPU Scheduling
 - Process Synchronization
 - IPC Mechanisms
 - Thread Management
 - Deadlock Concepts
 - Operating System Design
 - Data Visualization
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Author

Project developed by:

- chirombodesire
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GitHub Repository

Repository:

<https://github.com/chirombodesire7-ai/ASSIGNMENT-OS25311351006>